Lab 01 , Policy-Based Routing

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01. Objective and Topology

Policy-based routing lets a FortiGate forward traffic on something other than the destination address , source, incoming interface, protocol, port, type of service, or a known internet service. It sits above the routing table in the forwarding decision, which makes it powerful and easy to get wrong: an entry that matches will be used even when the routing table disagrees, and it carries no health state of its own.

This lab builds a two-upstream topology with a destination reachable through either path, establishes a baseline where the routing table has exactly one answer, and then proves two matching behaviors against it: steering by source subnet, and steering by protocol and port. Because the destination sits behind both ISPs, the path actually taken is visible without internet access.

1.2 Topology

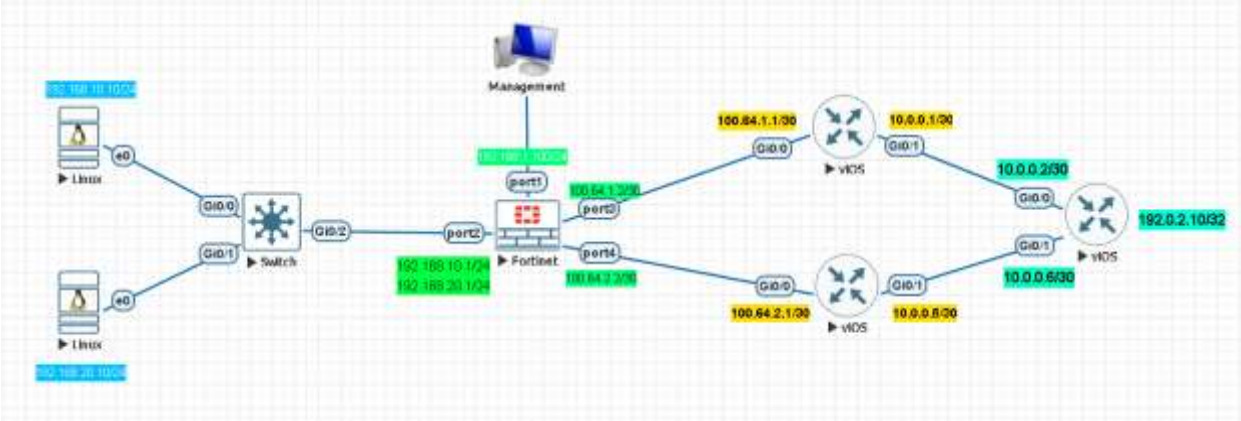


Figure 1 , Lab topology. Two upstream paths through ISP-A and ISP-B converge on INET, which hosts the test destination 192.0.2.10.

Device	Interface	Address	Connects to
FortiGate	port1	192.168.1.100/24	Management
FortiGate	port2	untagged, no IP	Switch Gi0/2 (802.1Q trunk)
FortiGate	vlan10 (USERS)	192.168.10.1/24	VLAN 10 on port2
FortiGate	vlan20 (OPS)	192.168.20.1/24	VLAN 20 on port2
FortiGate	port3 (WAN-A)	100.64.1.2/30	ISP-A Gi0/0
FortiGate	port4 (WAN-B)	100.64.2.2/30	ISP-B Gi0/0
ISP-A	Gi0/0 / Gi0/1	100.64.1.1/30 / 10.0.0.1/30	FortiGate port3 / INET

Device	Interface	Address	Connects to
ISP-B	Gi0/0 / Gi0/1	100.64.2.1/30 / 10.0.0.5/30	FortiGate port4 / INET
INET	Gi0/0 / Gi0/1 / Lo0	10.0.0.2 / 10.0.0.6 / 192.0.2.10	ISP-A / ISP-B / test destination
Linux-1	eth0	192.168.10.10/24	Switch Gi0/0, VLAN 10
Linux-2	eth0	192.168.20.10/24	Switch Gi0/1, VLAN 20

Why the destination sits behind both ISPs

A policy route can only be proven by observing which path traffic took. With a single destination reachable through either upstream, the first hop in a traceroute and the gateway recorded in the session table are direct evidence , no internet access and no public addressing required.

02. Access Switch

The two Linux hosts sit in separate VLANs behind one switch, and the FortiGate receives both over a single tagged link. This matters for the lab rather than being incidental: the ingress interface recorded by a policy route is the VLAN interface, not the physical port, and that distinction is the first thing to get wrong when writing the rule.

```
Switch>enable
Switch#conf
Switch#configure ter
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#no ip doma
Switch(config)#no ip domain-lookup
```

Figure 2 , Disabling DNS lookup first, so that a mistyped command fails immediately instead of stalling on name resolution.

```
Switch(config)#vlan 10
Switch(config-vlan)#name users
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name OPS
```

Figure 3 , Creating VLAN 10 (users) and VLAN 20 (OPS).

```
Switch(config)#interface gigabitEthernet0/0
Switch(config-if)#disc
Switch(config-if)#des
Switch(config-if)#description Linux-1 (VLAN 10)
Switch(config-if)#sw
Switch(config-if)#switchport mode
Switch(config-if)#switchport mode acc
Switch(config-if)#switchport mode access
Switch(config-if)#
*Nov  8 15:07:35.648: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (
g Wizard)
Switch(config-if)#sw
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vl
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no sh
Switch(config-if)#no shutdown
```

Figure 4 , Gi0/0 as an access port in VLAN 10, facing Linux-1.

```
Switch(config)#interface gigabitEthernet0/1
Switch(config-if)#des
Switch(config-if)#description Linux-2 (VLAN 20)
Switch(config-if)#sw
Switch(config-if)#switchport mode acc
Switch(config-if)#switchport mode access
Switch(config-if)#sw
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no sh
Switch(config-if)#no shutdown
```

Figure 5 , Gi0/1 as an access port in VLAN 20, facing Linux-2.

```
Switch(config)#interface gigabitEthernet0/2
Switch(config-if)#des
Switch(config-if)#description Trunk to FGT Port2
Switch(config-if)#sw
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk en
Switch(config-if)#switchport trunk encapsulation do
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#sw
Switch(config-if)#switchport mod
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Switch(config-if)#sw
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk all
Switch(config-if)#switchport trunk allowed val
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#no shu
Switch(config-if)#no shutdown
Switch(config-if)#end
```

Figure 6 , Gi0/2 configured as an 802.1Q trunk toward FortiGate port2, with VLANs 10 and 20 allowed.

```
Switch#write memory
Building configuration...
Compressed configuration from 3105 bytes to 1777 bytes[OK]
Switch#
*Nov  8 15:10:52.040: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*Nov  8 15:10:52.709: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
```

Figure 7 , Writing the running configuration to startup.

Verification

The switch is verified before the FortiGate is touched. A trunk that is not forwarding both VLANs produces a symptom on the FortiGate side that looks exactly like a FortiGate misconfiguration.

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gi0/3, Gi1/0, Gi1/1, Gi1/2
                                                Gi1/3
10   users                            active    Gi0/0
20   OPS                              active    Gi0/1
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```

Figure 8 , show vlan brief: VLAN 10 on Gi0/0, VLAN 20 on Gi0/1, both active.

```
Switch#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Gi0/2       on               802.1q         trunking      1

Port        Vlans allowed on trunk
Gi0/2       10,20

Port        Vlans allowed and active in management domain
Gi0/2       10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gi0/2       10,20
```

Figure 9 , show interfaces trunk: Gi0/2 trunking with 802.1Q, VLANs 10 and 20 allowed, active and forwarding.

03. ISP-A

ISP-A is the preferred upstream. Its only jobs are to terminate the transit link from the FortiGate, reach INET, and know how to return traffic.

```
Router>en
Router>enable
Router#con
Router#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hos
Router(config)#hostname ISP-A
```

Figure 10 , Setting the hostname.

```
ISP-A(config)#no ip domain-lookup
ISP-A(config)#line con 0
ISP-A(config-line)#lg
ISP-A(config-line)#log
ISP-A(config-line)#logg
ISP-A(config-line)#logging synch
ISP-A(config-line)#logging synchronous
ISP-A(config-line)#exec-time
ISP-A(config-line)#exec-timeout 0 0
ISP-A(config-line)#exit
```

Figure 11 , Console logging configuration, so that interface state messages do not interleave with typed input.

```
ISP-A(config)#interface gigabitEthernet0/0
ISP-A(config-if)#des
ISP-A(config-if)#description to FGT Port3
ISP-A(config-if)#ip addd
ISP-A(config-if)#ip addd
ISP-A(config-if)#ip add
ISP-A(config-if)#ip address 100.64.1.1 255.255.255.252
ISP-A(config-if)#no sh
ISP-A(config-if)#no shutdown
```

Figure 12 , Gi0/0, the transit link to FortiGate port3 (100.64.1.1/30).

```
ISP-A(config)#interface gig
ISP-A(config)#interface gigabitEthernet0/1
ISP-A(config-if)#des
ISP-A(config-if)#description to INET
ISP-A(config-if)#ip add
ISP-A(config-if)#ip address 10.0.0.1 255.255.255.252
ISP-A(config-if)#no sh
ISP-A(config-if)#no shutdown
```

Figure 13 , Gi0/1, the link toward INET (10.0.0.1/30).

```
ISP-A(config)#ip route 192.0.2.10 255.255.255.255 10.0.0.2
ISP-A(config)#ip route 192.168.10.0 255.255.255.0 10.64.1.2
ISP-A(config)#ip route 192.168.20.0 255.255.255.0 10.64.1.2
```

Figure 14 , Static routes: the test destination via INET, and the two internal subnets back toward the FortiGate.

The return routes are optional here

NAT is enabled on the FortiGate policies, so the internal addresses never appear upstream and the ISPs only ever see 100.64.1.2 and 100.64.2.2. The routes to 192.168.10.0/24 and 192.168.20.0/24 are configured anyway, so that NAT can be disabled later to observe real source addresses without re-configuring both upstreams.

04. ISP-B

ISP-B is identical to ISP-A apart from addressing. Keeping the two symmetrical means any difference in behaviour later is attributable to the FortiGate rather than to an upstream asymmetry.

```
ISP-B(config)#interface gigabitEthernet0/0
ISP-B(config-if)#des
ISP-B(config-if)#description to FGT Port4
ISP-B(config-if)#ip add
ISP-B(config-if)#ip address 100.64.2.1 255.255.255.252
ISP-B(config-if)#no sh
ISP-B(config-if)#no shutdown
```

Figure 15 , Gi0/0, the transit link to FortiGate port4 (100.64.2.1/30).

```
ISP-B(config)#interface gigabitEthernet0/1
ISP-B(config-if)#dess
ISP-B(config-if)#des
ISP-B(config-if)#description to INET
ISP-B(config-if)#ip add
ISP-B(config-if)#ip address 10.0.0.5 255.255.255.252
ISP-B(config-if)#no sh
ISP-B(config-if)#no shutdown
```

Figure 16 , Gi0/1, the link toward INET (10.0.0.5/30).

```
ISP-B(config)#ip route 192.0.2.10 255.255.255.255 10.0.0.6
ISP-B(config)#ip rout
ISP-B(config)#ip route 192.168.10.0 255.255.255.0 100.64.2.2
ISP-B(config)#ip ro
ISP-B(config)#ip route 192.168.20.0 255.255.255.0 100.64.2.2
```

Figure 17 , Static routes on ISP-B.

05. INET and the Test Service

INET is the shared destination. Its loopback is what both paths lead to, and it also hosts the HTTPS service used later to generate a TCP session on a known port while ICMP from the same host runs alongside it.

```
INET(config)#interface loopback0
INET(config-if)#ip add
INET(config-if)#ip address 192.0.2.10 255.255.255.255
```

Figure 18 , Loopback0 carrying the test destination 192.0.2.10/32.

```
INET(config)#interface gigabitEthernet0/0
INET(config-if)#des
INET(config-if)#description to ISP-A
INET(config-if)#ip add
INET(config-if)#ip address 10.0.0.2 255.255.2555.252
                                              ^
% Invalid input detected at '^' marker.

INET(config-if)#ip address 10.0.0.2 255.255.255.252
```

Figure 19 , Gi0/0 toward ISP-A. The first attempt was rejected for a mistyped mask and corrected immediately.

```
INET(config)#ip route 100.64.1.0 255.255.255.252 10.0.0.1
INET(config)#ip route 100.64.2.0 255.255.255.252 10.0.0.5
INET(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.1
INET(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.1
```

Figure 20 , Static routes back toward both transit networks.

5.1 HTTPS listener

```
INET(config)#ip http secure-server
% Generating 1024 bit RSA keys, keys will be non-exportable...
[OK] (elapsed time was 0 seconds)
Failed to generate persistent self-signed certificate.
    Secure server will use temporary self-signed certificate.
```

Figure 21 , Enabling the secure server. The persistent self-signed certificate could not be generated, so a temporary one was used instead.

```
INET#show ip http server status | include secure
HTTP secure server capability: Present
HTTP secure server status: Enabled
HTTP secure server port: 443
HTTP secure server ciphersuite:  3des-ede-cbc-sha des-cbc-sha rc4-128-m
HTTP secure server client authentication: Disabled
HTTP secure server trustpoint:
HTTP secure server active session modules: ALL
```

Figure 22 , show ip http server status: secure server enabled on port 443, confirming the warning did not prevent the service from starting.

```
INET(config)#username labadmin privilege 15 secret Lab12345
INET(config)#ip htt
INET(config)#ip http authe
INET(config)#ip http authentication local
INET(config)#end
```

Figure 23 , Creating the local user for HTTP authentication.

The certificate warning is not a failure for this lab. The test is TCP reachability on port 443 and which path the session takes, not certificate trust, and `curl -k` ignores the certificate entirely. The

practical consequence is that the temporary certificate is regenerated on every reload, so its fingerprint changes between sessions.

06. Linux Hosts

One host per VLAN, each with a static address and a default route pointing at its VLAN interface on the FortiGate.

```
eve@ubuntu:~$ ip addr flush dev eth0
Failed to send flush request: Operation not permitted
eve@ubuntu:~$ sudo ip addr flush dev eth0
[sudo] password for eve:
```

Figure 24 , Flushing the existing address. The operation requires root, so the command is repeated under sudo.

```
eve@ubuntu:~$ ip addr add 192.168.10.10/24 dev eth0
RTNETLINK answers: Operation not permitted
eve@ubuntu:~$ sudo ip addr add 192.168.10.10/24 dev eth0
eve@ubuntu:~$ ip link set eth0 up
RTNETLINK answers: Operation not permitted
eve@ubuntu:~$ sudo ip link set eth0 up
eve@ubuntu:~$ sudo ip route add default via 192.168.10.1
```

Figure 25 , Linux-1 addressed on VLAN 10: 192.168.10.10/24 with default gateway 192.168.10.1.

```
eve@ubuntu:~$ ip addr flush dev eth0
eve@ubuntu:~$ sudo ip addr add 192.168.20.10/24 dev eth0
[sudo] password for eve:
eve@ubuntu:~$ sudo ip link set eth0 up
eve@ubuntu:~$ ip route add default via 192.168.20.1
RTNETLINK answers: Operation not permitted
eve@ubuntu:~$ sudo ip route add default via 192.168.20.1
```

Figure 26 , Linux-2 addressed on VLAN 20: 192.168.20.10/24 with default gateway 192.168.20.1.

07. FortiGate Base Configuration

The base configuration is deliberately shaped so that the routing table has exactly one answer for any destination. Both defaults exist, but at different administrative distances, so only the WAN-A route is installed. Every path change observed later is therefore attributable to a policy route rather than to routing.

7.1 WAN interfaces

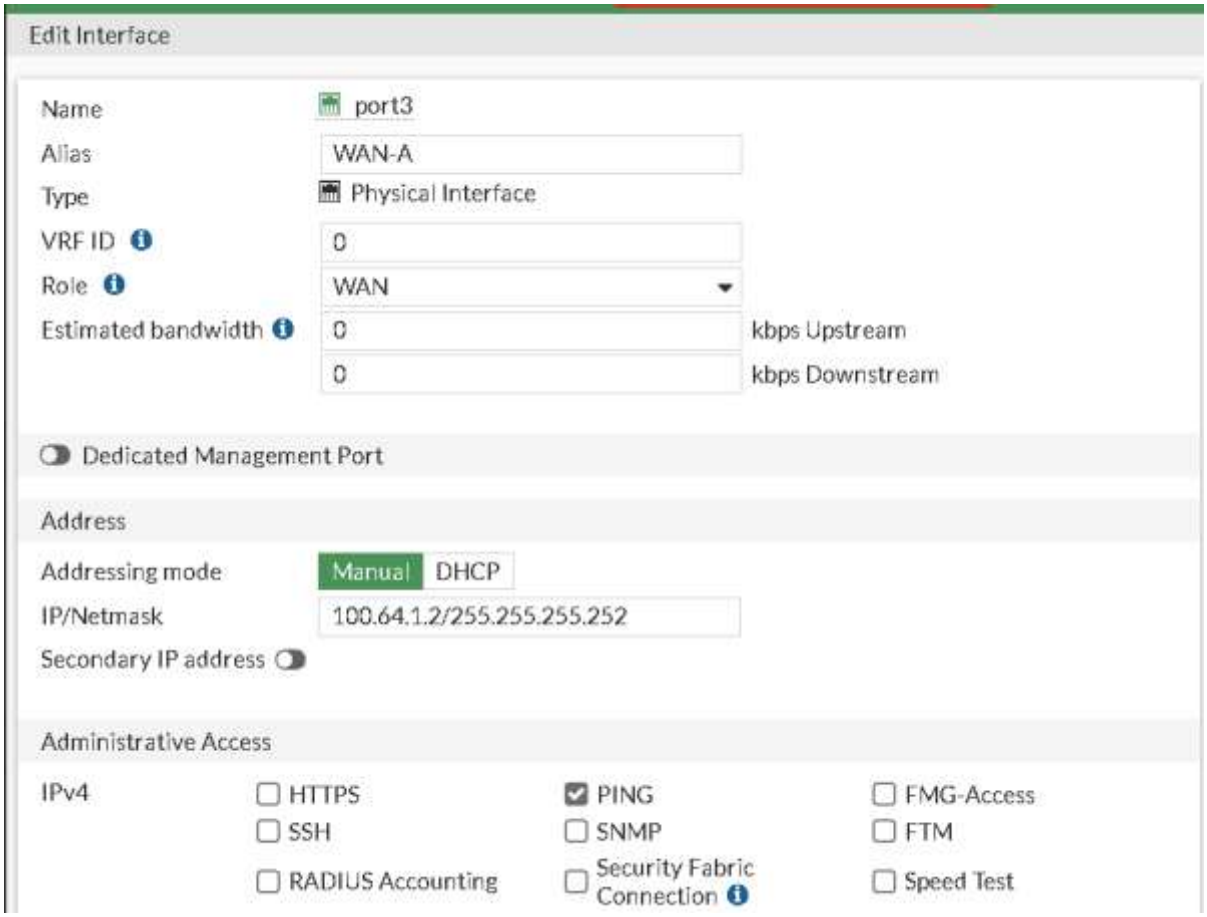


Figure 27 , port3 configured as WAN-A, 100.64.1.2/30.

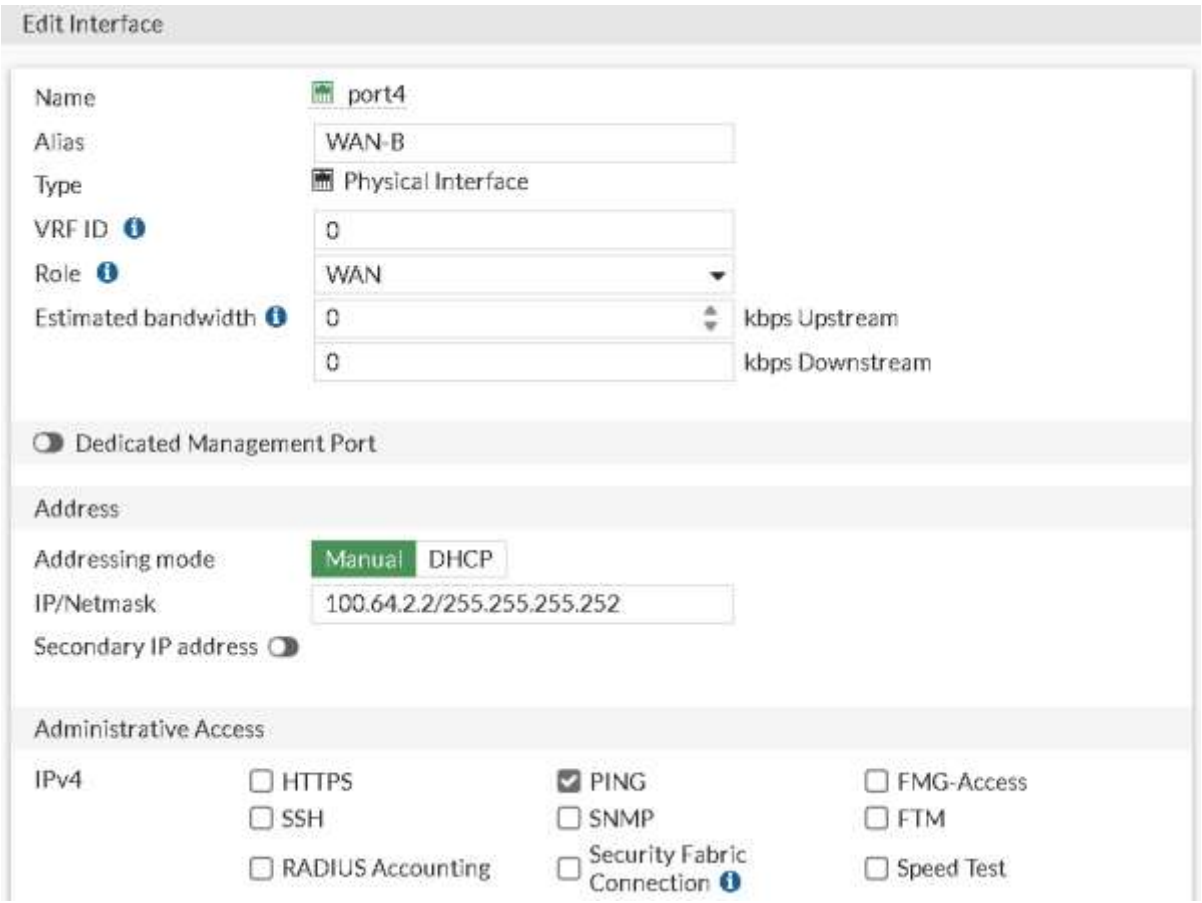


Figure 28 , port4 configured as WAN-B, 100.64.2.2/30.

7.2 Tagged LAN and VLAN interfaces

port2 carries tagged traffic only. It is given the LAN role and left without an address; the addressing lives on the VLAN interfaces built on top of it.

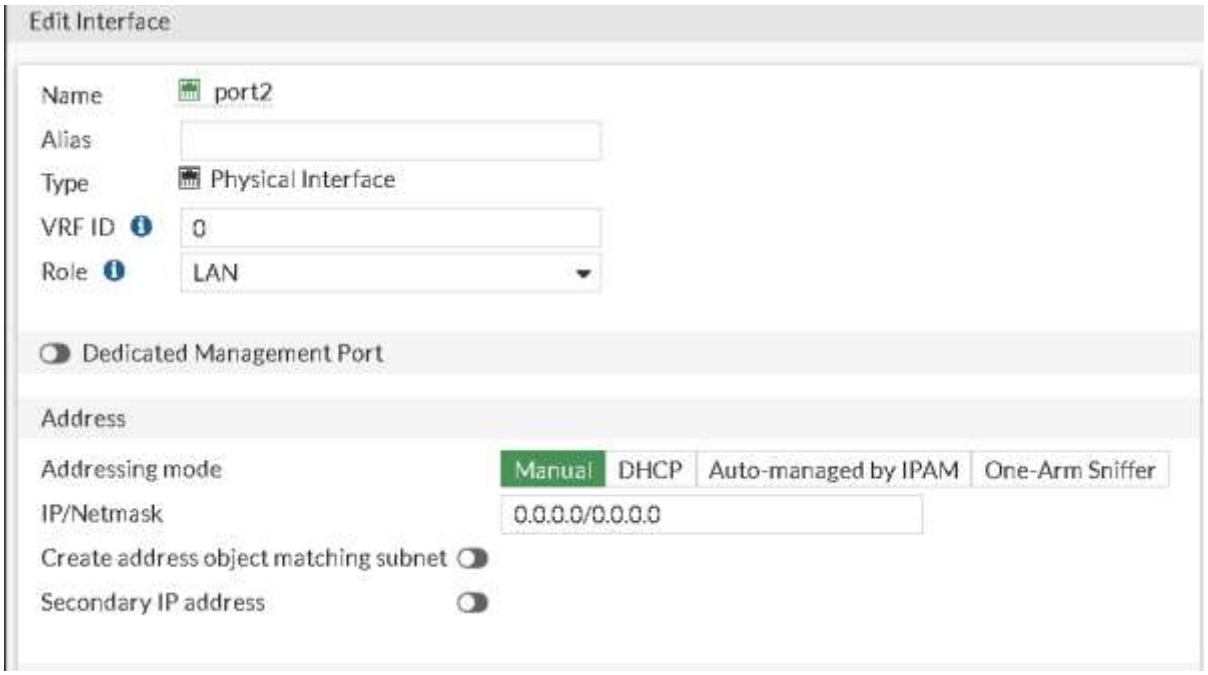


Figure 29 , port2 as the tagged trunk toward the switch, carrying no address of its own.

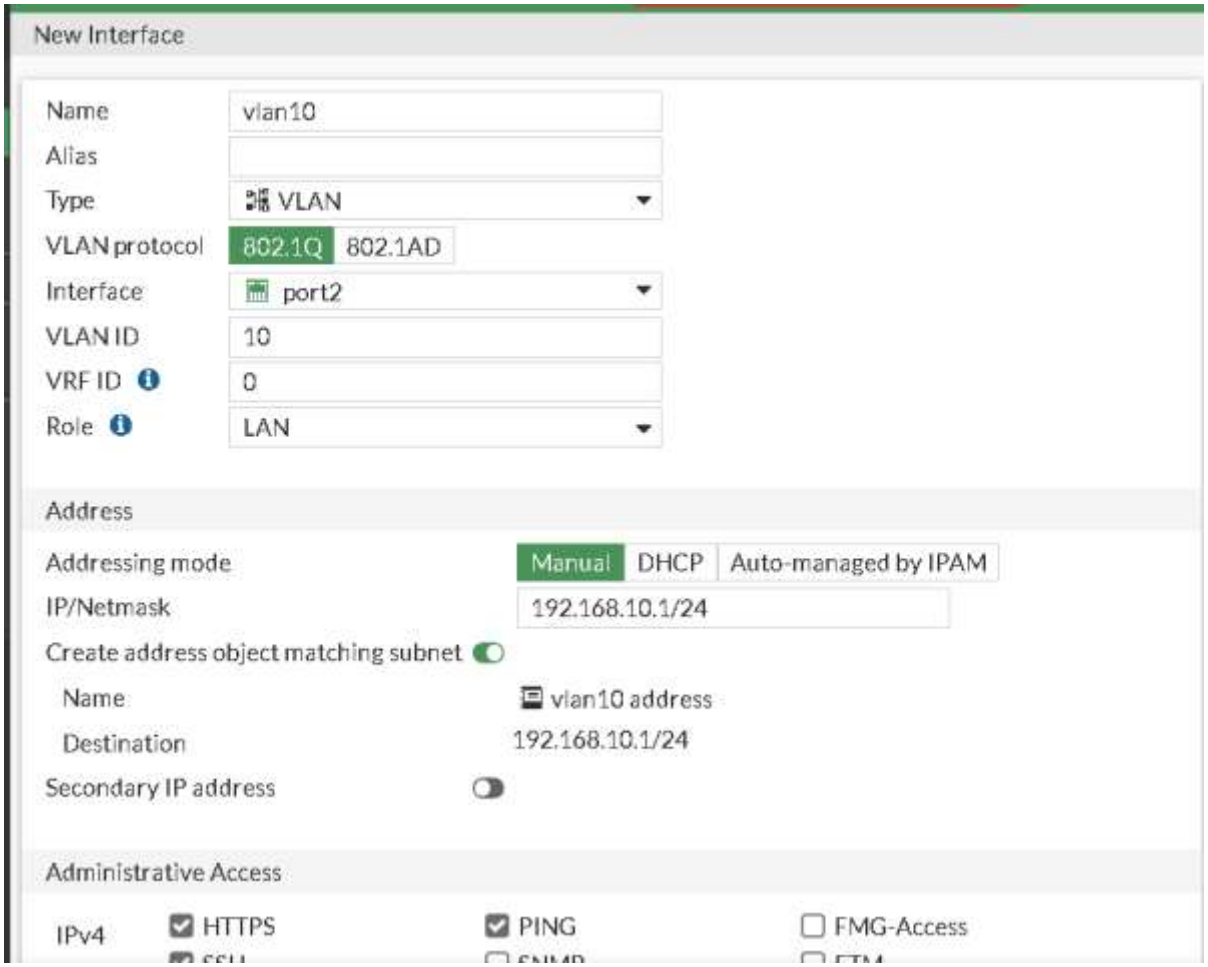


Figure 30 , The two VLAN interfaces: vlan10 (192.168.10.1/24) and vlan20 (192.168.20.1/24), both on port2.

7.3 Address objects

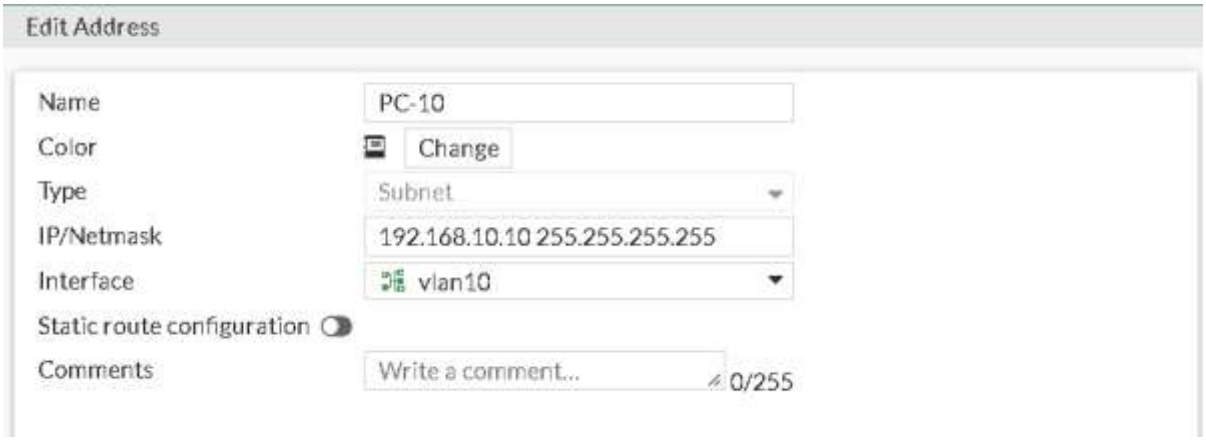


Figure 31 , Address objects for the two VLAN subnets and the INET destination, used later by the policy routes and policies.

7.4 Static routes



Figure 32 , Default route via WAN-A, administrative distance 10.



Figure 33 , Default route via WAN-B, administrative distance 20 , present but not installed.

7.5 Firewall policies

Two policies, one per egress. A policy route selects the exit interface, but the firewall still has to permit the traffic on that interface. Keeping the two policies separate rather than using a zone also means the forward traffic log names the path directly, which is what makes the later evidence readable.

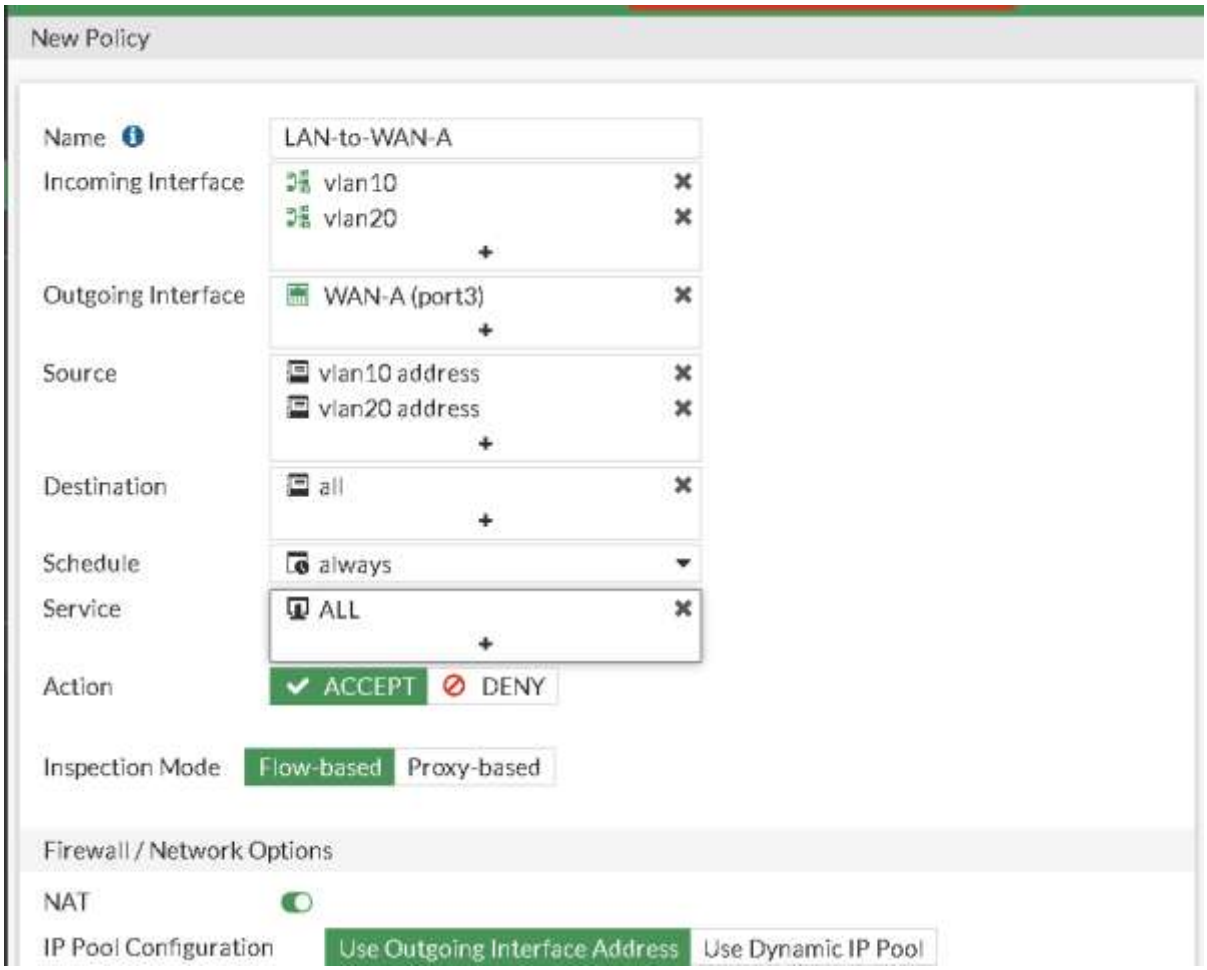


Figure 34 , LAN-to-WAN-A: both VLAN interfaces to port3, NAT enabled.

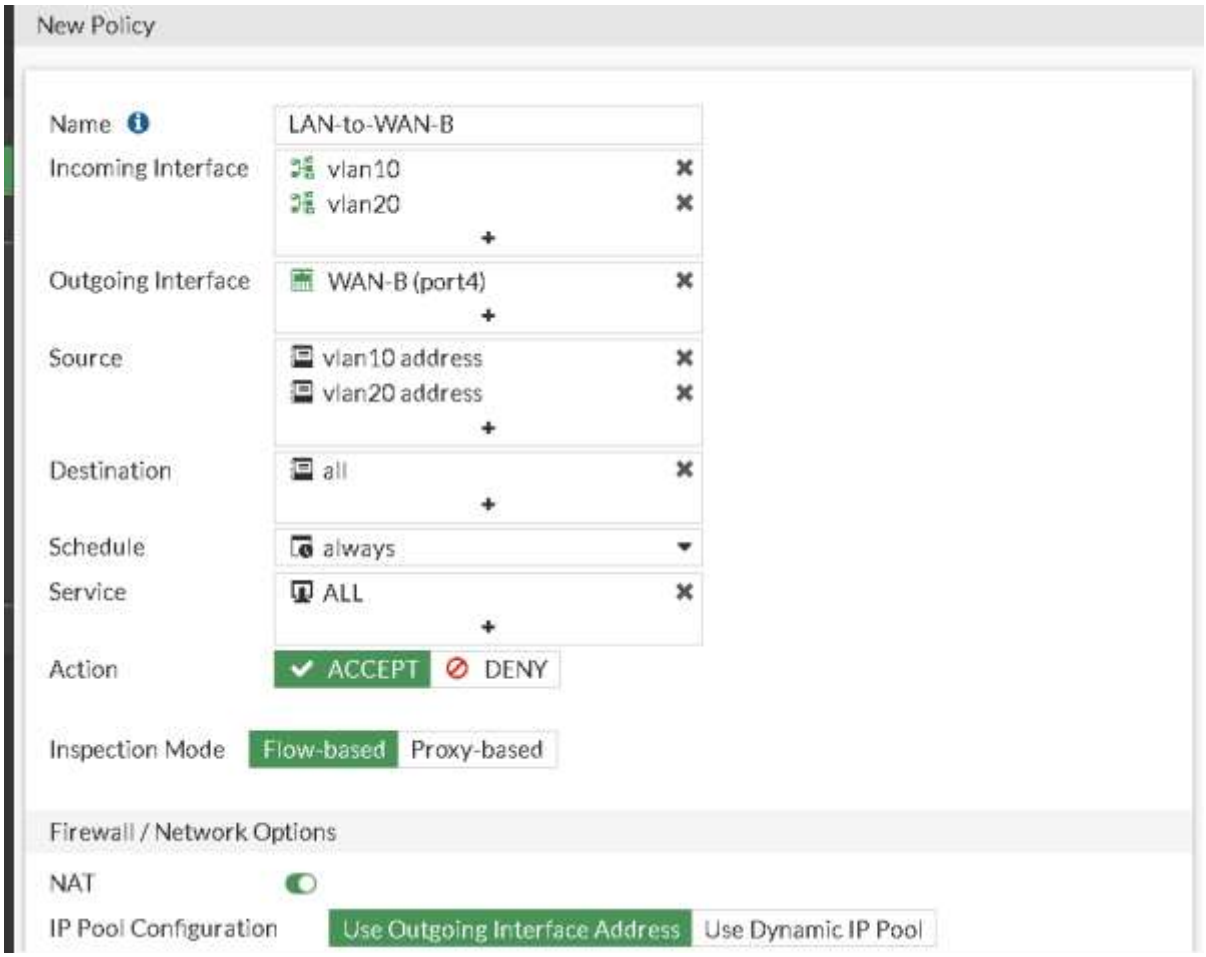


Figure 35 , LAN-to-WAN-B: the same sources to port4, NAT enabled. Without this policy a correct policy route would silently fail.

08. Baseline

The baseline is captured before any policy route exists. Every later observation is compared against it, and a baseline taken afterwards would be worthless.

```
FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default

Routing table for VRF=0
S*      0.0.0.0/0 [10/0] via 100.64.1.1, port3, [1/0]
C       100.64.1.0/30 is directly connected, port3
C       100.64.2.0/30 is directly connected, port4
C       192.168.1.0/24 is directly connected, port1
C       192.168.10.0/24 is directly connected, vlan10
C       192.168.20.0/24 is directly connected, vlan20
```

Figure 36 , The routing table: a single default, 0.0.0.0/0 [10/0] via 100.64.1.1 on port3. The WAN-B route is configured but not installed.

```
FortiGate-VM64-KVM # diagnose firewall proute list
list route policy info(vf=root):

FortiGate-VM64-KVM #
```

Figure 37 , diagnose firewall proute list returns nothing. No policy routes exist yet.

```
eve@ubuntu:~$ ping 192.0.2.10 -c 4
PING 192.0.2.10 (192.0.2.10) 56(84) bytes of data.
64 bytes from 192.0.2.10: icmp_seq=1 ttl=253 time=5.77 ms
64 bytes from 192.0.2.10: icmp_seq=2 ttl=253 time=4.57 ms
64 bytes from 192.0.2.10: icmp_seq=3 ttl=253 time=2.55 ms
64 bytes from 192.0.2.10: icmp_seq=4 ttl=253 time=2.85 ms
```

Figure 38 , Linux-1 (VLAN 10) reaching 192.0.2.10.

```
eve@ubuntu:~$ ping 192.0.2.10 -c 4
PING 192.0.2.10 (192.0.2.10) 56(84) bytes of data.
64 bytes from 192.0.2.10: icmp_seq=1 ttl=253 time=4.31 ms
64 bytes from 192.0.2.10: icmp_seq=2 ttl=253 time=4.79 ms
64 bytes from 192.0.2.10: icmp_seq=3 ttl=253 time=2.69 ms
64 bytes from 192.0.2.10: icmp_seq=4 ttl=253 time=3.56 ms
```

Figure 39 , Linux-2 (VLAN 20) reaching the same destination.

Date/Time		Source	Device	Destination	Application Name	Result	Policy ID
2026/11/08 12:11:24		192.168.10.10	00:50:00:00:03:00	192.0.2.10		✔ 336 B / 336 B	LAN-to-WAN-A (1)
2026/11/08 12:10:59		192.168.20.10	00:50:00:00:04:00	192.0.2.10		✔ 336 B / 336 B	LAN-to-WAN-A (1)

Figure 40 , Forward traffic log: both hosts matched LAN-to-WAN-A (1). Nothing is using WAN-B.

Both subnets reach the destination, and every session is on the WAN-A policy , the routing table is the only thing making the decision at this point. A configuration backup was taken here, before the tests begin.

09. Source-Based Policy Routing

The first test moves everything from VLAN 20 onto WAN-B while VLAN 10 continues to follow the routing table. This is the simplest form of policy routing and it proves the central point: a matching policy route overrides the installed route.

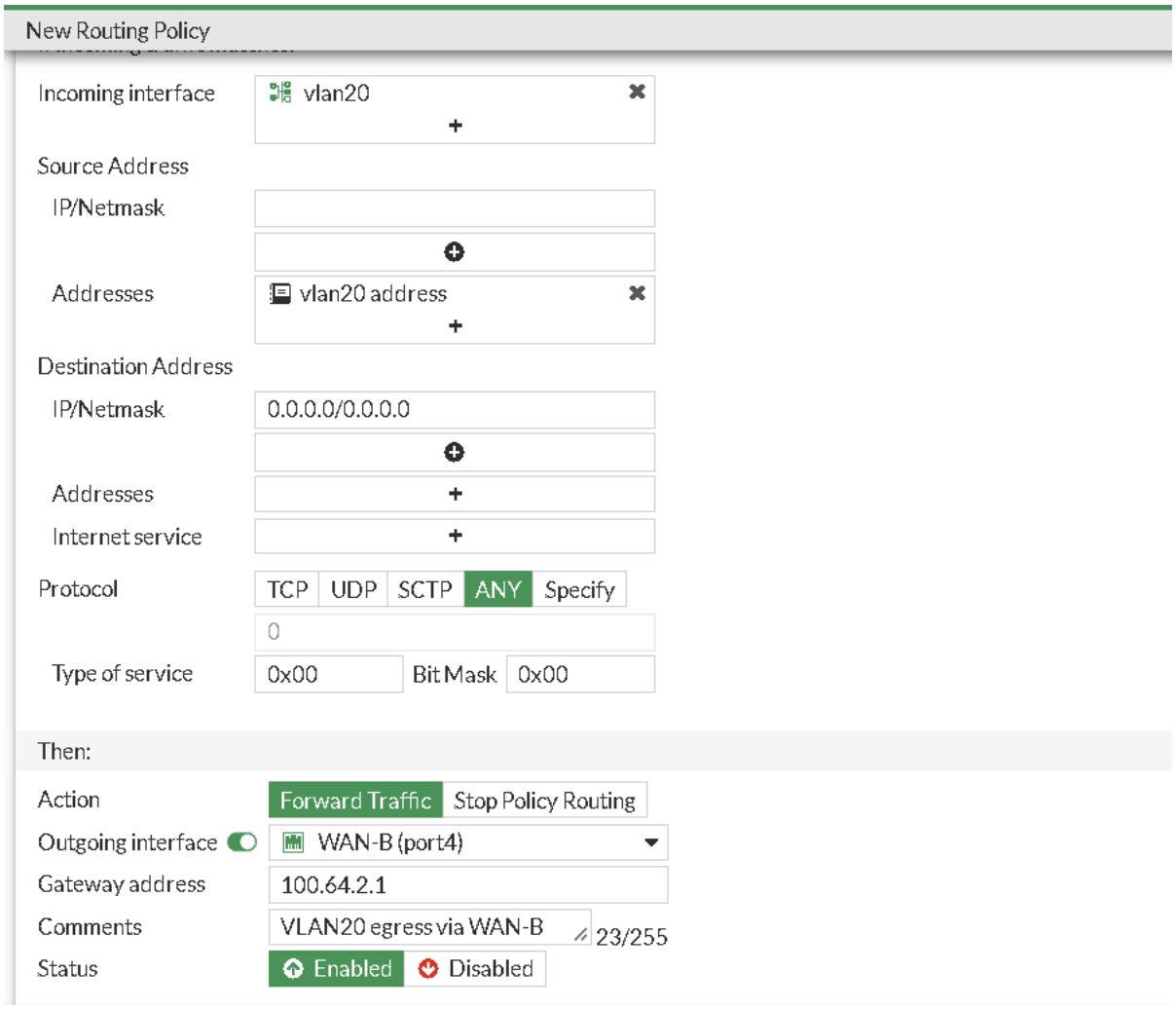


Figure 41 , The policy route: incoming interface vlan20, source the VLAN 20 subnet, any destination, forwarded out WAN-B (port4) with gateway 100.64.2.1.

What the device recorded

```
FortiGate-VM64-KVM # diagnose firewall proute list
list route policy info(vf=root):

id=1 dscp_tag=0xff 0xff flags=0x0 tos=0x00 tos_mask=0x00 protocol=0 sport=0-0 iif=16 dport=0-65535 path(1) oif=6(port4) gwy=100.64.2.1
source(1): 192.168.20.0-192.168.20.255
destination wildcard(1): 0.0.0.0/0.0.0.0
hit_count=0 last_used=2026-11-08 13:35:08
```

Figure 42 , diagnose firewall proute list showing the entry: iif=16 (vlan20), oif=6 (port4), gwy=100.64.2.1, source 192.168.20.0–192.168.20.255, destination wildcard.

Field in the output	Meaning
id=1	Sequence number , also the evaluation order. First match wins; there is no longest-match logic.
iif=16	Ingress interface index , vlan20, not port2. Policy routes see the VLAN interface.
oif=6(port4)	Egress chosen by the rule.
gwy=100.64.2.1	Next hop the FortiGate will ARP for. Without it the entry matches and the traffic is dropped on an Ethernet segment.
protocol=0, sport/dport full range	No protocol or port narrowing , this entry matches everything from the subnet.
hit_count / last_used	Counters for the entry. Read immediately after creation they say nothing useful.

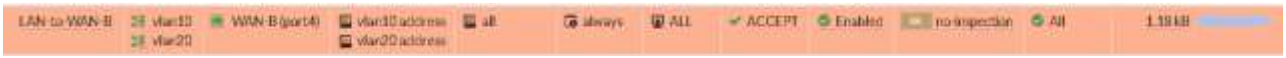


Figure 43 , The LAN-to-WAN-B policy now carrying traffic (1.18 kB). Before the policy route existed it had never been matched.

Gateway address is not optional

An entry with an outgoing interface but no gateway will match traffic and then drop it silently on a broadcast segment, because the FortiGate has no next hop to resolve. The symptom is indistinguishable from a routing failure, which is why the gateway field is filled in every entry in this lab.

10. Protocol and Port Matching

The second test narrows the match: only HTTPS from Linux-1 leaves through WAN-B, while everything else from the same host stays on WAN-A. Two concurrent sessions from one source on two different paths is the clearest demonstration in the lab that policy routing is evaluated per session, not per host.

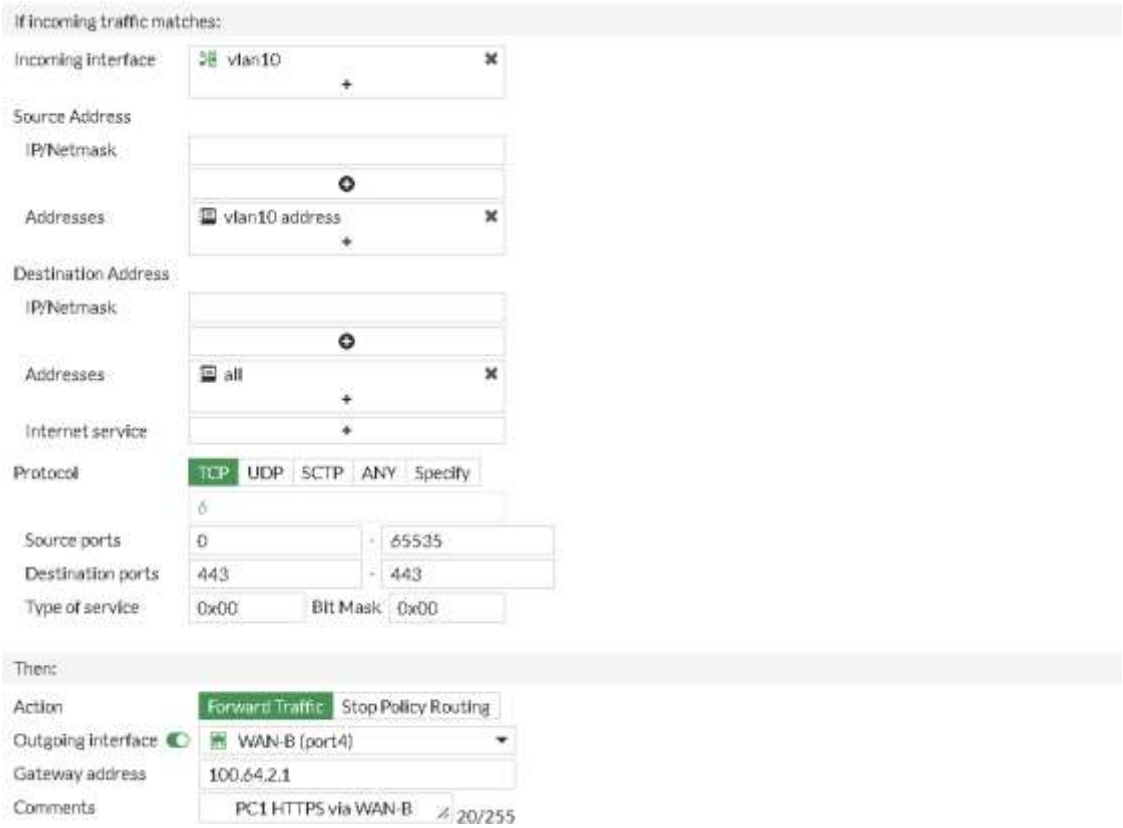


Figure 44 , The second policy route: incoming interface vlan10, source the VLAN 10 address object, protocol TCP, destination ports 443–443, forwarded out WAN-B.

Generating both flows at once

```
eve@ubuntu:~$ while true; do curl -k -s -o /dev/null https://192.0.2.10/; sleep
1; done
```

Figure 45 , A curl loop against the HTTPS listener, producing a continuous TCP/443 session from Linux-1.

```
eve@ubuntu:~$ ping 192.0.2.10
PING 192.0.2.10 (192.0.2.10) 56(84) bytes of data.
64 bytes from 192.0.2.10: icmp_seq=1 ttl=253 time=5.37 ms
64 bytes from 192.0.2.10: icmp_seq=2 ttl=253 time=2.85 ms
64 bytes from 192.0.2.10: icmp_seq=3 ttl=253 time=2.38 ms
64 bytes from 192.0.2.10: icmp_seq=4 ttl=253 time=2.90 ms
64 bytes from 192.0.2.10: icmp_seq=5 ttl=253 time=2.76 ms
64 bytes from 192.0.2.10: icmp_seq=6 ttl=253 time=2.88 ms
64 bytes from 192.0.2.10: icmp_seq=7 ttl=253 time=2.75 ms
64 bytes from 192.0.2.10: icmp_seq=8 ttl=253 time=2.86 ms
64 bytes from 192.0.2.10: icmp_seq=9 ttl=253 time=2.64 ms
64 bytes from 192.0.2.10: icmp_seq=10 ttl=253 time=2.89 ms
64 bytes from 192.0.2.10: icmp_seq=11 ttl=253 time=2.61 ms
64 bytes from 192.0.2.10: icmp_seq=12 ttl=253 time=3.68 ms
64 bytes from 192.0.2.10: icmp_seq=13 ttl=253 time=2.81 ms
64 bytes from 192.0.2.10: icmp_seq=14 ttl=253 time=2.90 ms
64 bytes from 192.0.2.10: icmp_seq=15 ttl=253 time=3.41 ms
64 bytes from 192.0.2.10: icmp_seq=16 ttl=253 time=3.44 ms
64 bytes from 192.0.2.10: icmp_seq=17 ttl=253 time=2.89 ms
```

Figure 46 , A ping running from the same host at the same time, providing the ICMP flow for comparison.

The session table

```
FortiGate-VM64-KVM # diagnose sys session filter src 192.168.10.10

FortiGate-VM64-KVM # diagnose sys session list

session info: proto=6 proto_state=05 duration=0 expire=0 timeout=3600 flags=00000000 socktype=0 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id=0 policy_dir=0 tunnel=/ vlan_cos=0/255
state=log may_dirty f00
statistic(bytes/packets/allow_err): org=1310/10/1 reply=1640/9/1 tuples=2
tx speed(Bps/kbps): 0/0 rx speed(Bps/kbps): 0/0
orgin->sink: org pre->post, reply pre->post dev=15->6/6->15 gwy=100.64.2.1/192.168.10.10
hook=post dir=org act=snat 192.168.10.10:39934->192.0.2.10:443(100.64.2.2:39934)
hook=pre dir=reply act=dnat 192.0.2.10:443->100.64.2.2:39934(192.168.10.10:39934)
pos/(before,after) 0/(0,0), 0/(0,0)
src_mac=00:50:00:00:03:00
misc=0 policy_id=2 pol_uuid_idx=14730 auth_info=0 chk_client_info=0 vd=0
serial=000018db tos=ff/ff app_list=0 app=0 url_cat=0
route_policy_id=2
rpdb_link_id=00000002 ngfwid=n/a
npu_state=0x000100

session info: proto=1 proto_state=00 duration=33 expire=59 timeout=0 flags=00000000 socktype=0 sockport=0 av_idx=0 use=3
origin-shaper=
reply-shaper=
per_ip_shaper=
class_id=0 ha_id=0 policy_dir=0 tunnel=/ vlan_cos=0/255
state=log may_dirty f00
statistic(bytes/packets/allow_err): org=2772/33/1 reply=2772/33/1 tuples=2
tx speed(Bps/kbps): 83/0 rx speed(Bps/kbps): 83/0
orgin->sink: org pre->post, reply pre->post dev=15->5/5->15 gwy=100.64.1.1/192.168.10.10
hook=post dir=org act=snat 192.168.10.10:3489->192.0.2.10:8(100.64.1.2:63905)
```

Figure 47 , Two sessions from 192.168.10.10. The TCP session uses dev=15->6 with gwy=100.64.2.1 and carries route_policy_id=2; the ICMP session uses dev=15->5 with gwy=100.64.1.1.

Session field	TCP/443 session	ICMP session
Protocol	proto=6	proto=1
Interfaces	dev=15->6 (vlan10 to port4)	dev=15->5 (vlan10 to port3)
Gateway	gwy=100.64.2.1 (WAN-B)	gwy=100.64.1.1 (WAN-A)

Session field	TCP/443 session	ICMP session
Source NAT	act=snat to 100.64.2.2	act=snat to 100.64.1.2
Policy	policy_id=2 (LAN-to-WAN-B)	LAN-to-WAN-A
Policy route	route_policy_id=2	none , routed normally

The `route_policy_id` field is the single most direct piece of evidence in this lab: it names the policy route that decided the session, on the session itself. The ICMP session carries no such field, which is what distinguishes a policy-routed flow from a routed one.

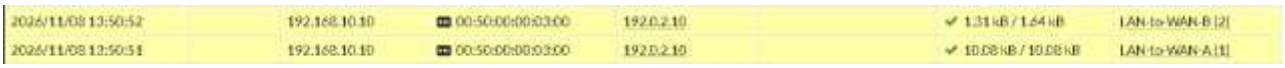


Figure 48 , Forward traffic log: the same source, 192.168.10.10, on LAN-to-WAN-B (2) and LAN-to-WAN-A (1) one second apart.

12. Command Reference

Command	What it answers
`get router info routing-table all`	What is actually installed, and at what distance
`diagnose firewall proute list`	Which policy routes exist, in evaluation order, with their match criteria
`diagnose netlink interface list`	Interface index to name mapping, needed to read the session table
`diagnose sys session filter src <ip>`	Narrow the session table to one host
`diagnose sys session list`	Per-session egress (dev), gateway (gwy), NAT action and route_policy_id
`diagnose debug flow trace start <n>`	Per-packet path through policy, policy route and route lookup
`show router policy`	Read back what the GUI wrote